

Further, landing at night time on a very short runway of a minimally-lit carrier, which is pitching, rolling, yawing and sliding, is nerve shattering and a mortal challenge. An error caused due to the pilot's judgement may cost not only his or her life but many other lives and also many aircraft.

Previous generations of pilots had only two optical clues to align the aircraft with respect to the carrier and then land safely. The first was called Fresnel Lens Optical Landing System (FLOLS) for vertical positioning of the aircraft. The second was called Centreline Lights for horizontal positioning of the aircraft. But modern-day pilots have an electronic aid called Precision Approach Landing System (PALS).

Precision Approach Landing System

PALS is really a pal for naval pilots. It has three modes: instrumented carrier landing system (ICLS) mode, automatic carrier landing system (ACLS) mode and talk-down mode.

ICLS mode. For an aircraft to land, it has to maintain a constant recommended speed, while losing altitude gradually, until touchdown on the runway. How it loses altitude with respect to the runway is called glideslope. For a runway on ground, the glideslope never changes but since the carrier rocks up and down, the glideslope also varies accordingly and is never constant. So a mere pitch up of the flight deck is drastic for the landing aircraft. Pilot of the aircraft must be made aware of this changing glideslope. This is done through a system called ICLS centred on a radar system called AN/SPN-41.

Though technically not an exact radar system, AN/SPN-41 engages in one-way transmission to the landing aircraft in Ku-band. It has a set of radars: a centreline unit and a flight deck unit. The former is situated beneath the centreline of the runway and the latter is situated slightly high on the flight-deck.

Using sector scanning, both units acquire and track the approaching aircraft. The flightdeck unit and centreline unit respectively measure the aircraft's vertical and horizontal deviation from the glideslope. Then AN/SPN-41 broadcasts the positional difference in pulse mode in Ku-band in real-time.

A system called aircraft approach control system (ACCS) present in the aircraft receives these signals. It extracts the positional information of the aircraft with respect to the glideslope from these signals. It then converts these signals as a set of horizontal and vertical lines on the pilot's HUD.

The vertical line indicates the position of the aircraft with respect to the centreline of the runway. It appears in the direction that the aircraft has to move so as to align with the centreline of the runway. The horizontal line indicates how much the aircraft must gain or lose altitude to fit in the glideslope.

When these two lines form a crosshair, the aircraft can be perfectly aligned with the glideslope. The pilot, after aligning with the glideslope, can gently nudge his controls to stay in the pitching glideslope until touchdown.

Talk-down mode. In this mode, the pilot is instructed by landing signal officers (LSOs). They work from an LSO console situated at the edge of the flight deck. This console gathers all flight data of the approaching aircraft from many systems. The LSO console computes the required flight path for landing, monitors the flight path of aircraft and the difference that the pilot has to cover. It takes inputs from the integrated launch and recovery television system (ILARTS), AN/SPN-44 radar and wind-measurement system.

ILARTS. This is a camera-and-television system with all-weather capability. In simple words, its cameras can videograph both during day and night with equal clarity. ILARTS has remotely-operated cameras at many






PG08

PG08 (Max. 10A)

- Power (W)
- Power Factor (PF)
- Current (mA)
- Voltage (V)
- Frequency (Hz)
- Total Usage Time (min)
- Total Used Energy (KWh)
- Carbon Emission (CO₂)

PG09H

PG09H (Max. 15A)

- TRMS Voltage (V)
- TRMS Current (mA)
- Power (W)
- Energy Usage Time (EUT)
- Energy Consumption (KWh)

Meco Meters Private Limited

Plot No. EL-80, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2787 3300 (Board), 2787 3311-16 (Sales)
Fax : 0091-22-2787 3310, 2787 3330
Email : sales@mecoinst.com
Web : www.mecoinst.com